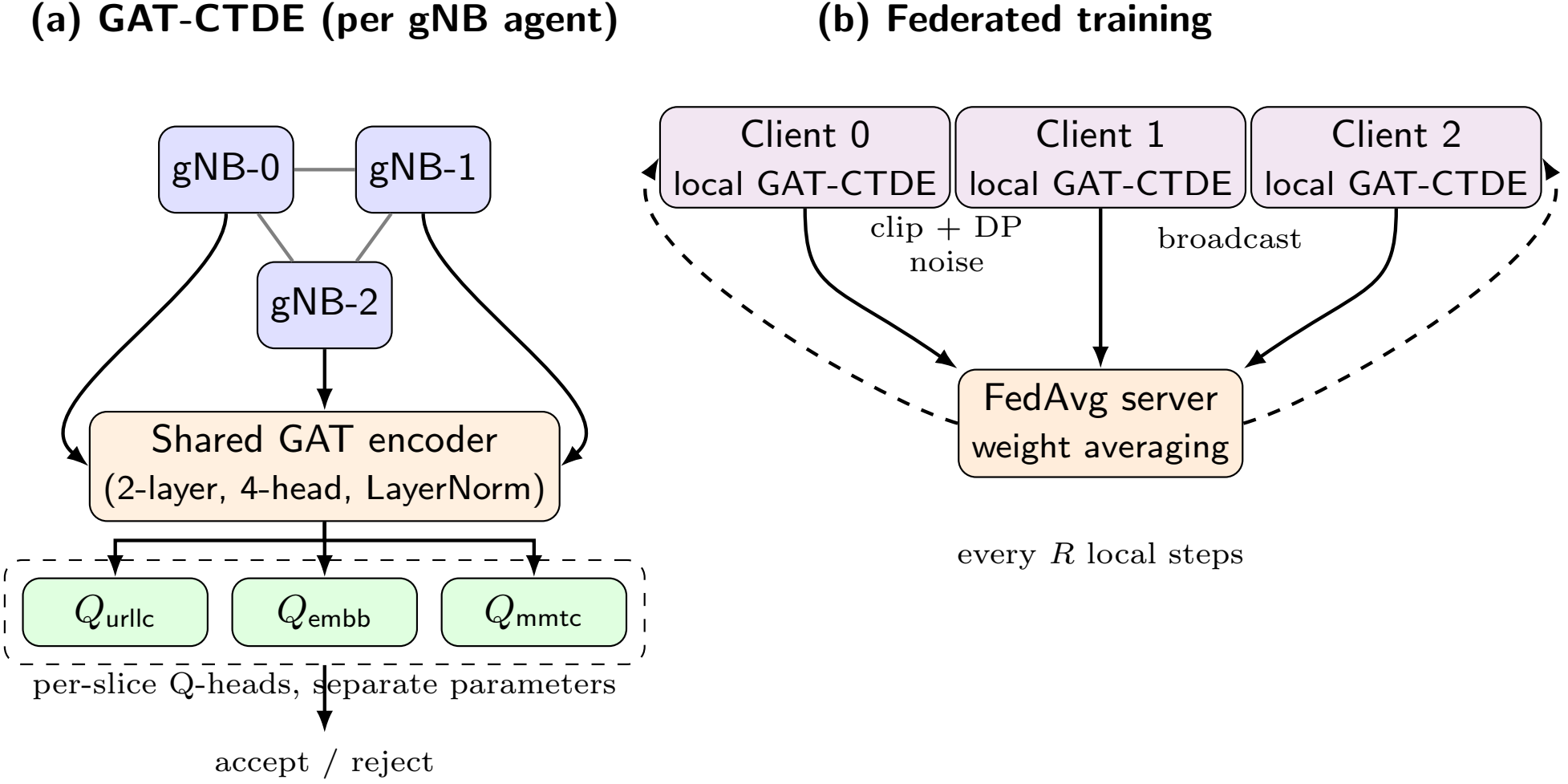

(a) GAT-CTDE (per gNB agent)
gNB-0
gNB-1
gNB-2
Shared GAT encoder
(2-layer, 4-head, LayerNorm)
Q_{urllc}
Q_{embb}
Q_{mmtc}
per-slice Q-heads, separate parameters
accept / reject
(b) Federated training
Client 0
local GAT-CTDE
Client 1
local GAT-CTDE
Client 2
local GAT-CTDE
clip + DP
noise
broadcast
FedAvg server
weight averaging
every R local steps